SUNGHYUN CHANG

Senior Software Engineer

INFO

Preferred Name
Sam

Location
Lansing, MI, USA

Contact
410-443-6161

Email
samashe.chang@gmail.com

LinkedIn
linkedin.com/in/samashe

Web-Portfolio
shsstudios.dev

GitHub
github.com/DobbyLikesCoding

LANGUAGES

Korean — Native

English — Professional working proficiency

SKILLS

Languages
C#, .NET Core / Framework, Python, C++, TypeScript

Desktop / Web
WPF, WinForms, ASP.NET Core, React

Architecture
Microservices, Distributed Systems, Real-Time Processing

Data & Integration
SQL Server, MariaDB, MySQL, Redis, TCP/IP, FIX

Messaging / DevOps
Kafka, Pulsar, Docker, Kubernetes, Jenkins, AWS

STRENGTHS

- Problem-First Engineering
- Software Architecture
- Root Cause Analysis
- Technical Leadership
- Operational Reliability

WORK AUTHORIZATION

U.S. permanent resident; no sponsorship required.

SUMMARY

Senior Software & Systems Engineer with 13+ years of experience across AI-driven fintech, EV battery manufacturing, and aerospace/defense. Expertise in C#/.NET, distributed and real-time systems, software architecture, and practical automation. Led an 8-person engineering team at QRAFT and now designs operational software for LG Energy Solution.

WORK EXPERIENCE

Senior Smart Factory Engineer

LG Energy Solution Michigan May 2025 – Present

- Lead technical architecture for MON's Club, an integrated manufacturing monitoring and automation initiative; develop solutions with C#/.NET, WPF, and MariaDB.
- Designed and developed VIMON (vision inspection monitoring) and CELLMON (cell tracking and PLC-log investigation) within MON's Club.
- Defined ROLLMON's initial concept, architecture, and database; application implementation was completed by another engineer.
- Investigate traceability and production issues through root-cause analysis to improve operational visibility and reliability.

Smart Factory Engineer II

Ultium Cells LLC May 2024 – May 2025

- Supported EV battery traceability deployment and stabilization, including Electrode Roll Map and Assembly Cell Tracking; coordinated testing, validation, and troubleshooting.

AXE Team Leader & Senior Software Engineer

QRAFT Technologies Dec 2022 – Nov 2023

- Led an 8-member cross-functional team (3 developers, 4 researchers, 1 QA) supporting AI-integrated institutional trading execution.
- Owned technical architecture, low-latency TCP/OMS integration, deployment coordination, and production reliability for AXE.

Senior Software Development Engineer

QRAFT Technologies Nov 2016 – Dec 2022

- Built C#/.NET microservices and real-time order processing components using TCP/IP, Redis, and message-based systems.
- Developed monitoring, CI/CD, and performance tools; supported client integration and trading platform operations.

Software Engineer

Pixoneer Geomatics Oct 2014 – Jun 2016

- Developed aerospace and defense software, including helicopter condition monitoring and air-traffic-related systems, across the SDLC.

Broadcast Technician (Contract)

MBC Broadcasting May 2009 – May 2011

- Maintained broadcast and studio equipment to support stable live operations and technical troubleshooting.

EDUCATION

Hanseo University B.E., Aerospace Software Engineering (2015), GPA 3.99/4.5

North-West University Exchange Student, South Africa (2013) - CS & Electronics